Replacement Parts Demand Forecasting

HORIZON HOBBY

Meet Team Nexus

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AGENDA

1. Existing process of forecasting & its impacts
2. Our model and its results
3. Recommendations

APPROACH

1. Logical & Realistic
2. Simple
3. Use AI to do all the grunt work

HORIZON'S CURRENT FORECASTING PROCESS

Bill of Materials

Assign Attachment Rate

- Flat 5-7% Rate

1 Year commitment

- ⬆ or ⬇
- Based on historical benchmarks

Sales Live

Biased Forecast

RECOMMENDED

(Feedback Loop Missing)

HORIZON HOBBY V/S TEAM NEXUS

$$\text{wMAPE} = \frac{\sum \mid Actual - Predicted \mid}{\sum \mid Actual \mid}$$

Horizon's wMAPE	Our wMAPE
61.9%	24.9%

FORECASTING BIAS

Part	Error	Outcome
Suspension Arm Set	+398%	Stock-out month 2
Hub & Spindle Set	+379%	Stock-out at launch
Front Tire Set	+220%	Biggest volume miss
Battery Tray	−80%	80% never sold

FORECASTING IMPACTS

Overall Forecast error	Parts under-forecast	Unmet Demand	Gross margin lost
61.9%	27/39	11,526	57.6%
wMAPE – Sprint Car	69% of all parts	units – lost forever	On every unfilled unit

Under-Forecast Failure Chain:

Root Cause
Flat 5% rate assigned to crash-critical parts

Stock-out
Wear parts depleted within weeks of launch

Customer Impact
Car sits broken. cannot race or enjoy hobby

Business Impact
C57.6% margin lost. SLA missed. Brand Damaged

Over-Forecast Failure Chain:

Root Cause
Same 5% rate applied to structural parts

Dead stock
Chassis & motor plates sit fo 12+ months

Capital Lock
$9,770 per product. $5.1M portfolio-wide

Business Impact
Less cash to buy parts that actually sell

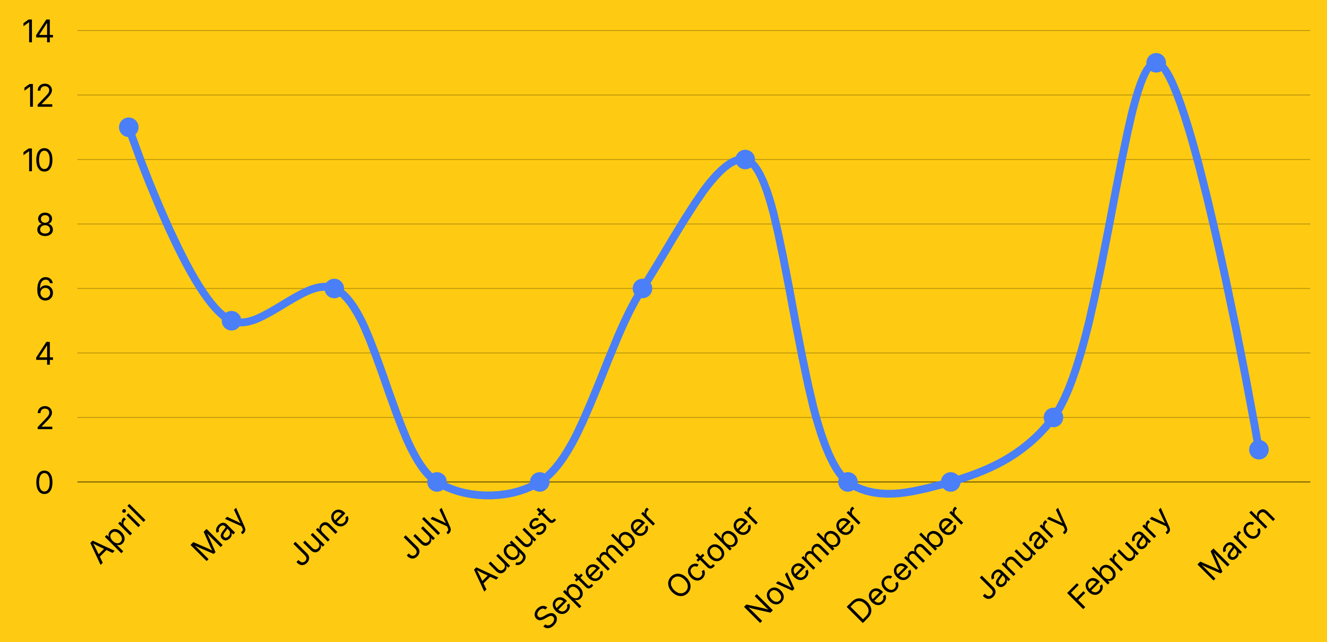
WHY IS THE FORECAST WRONG?
HOW TO FIX IT?

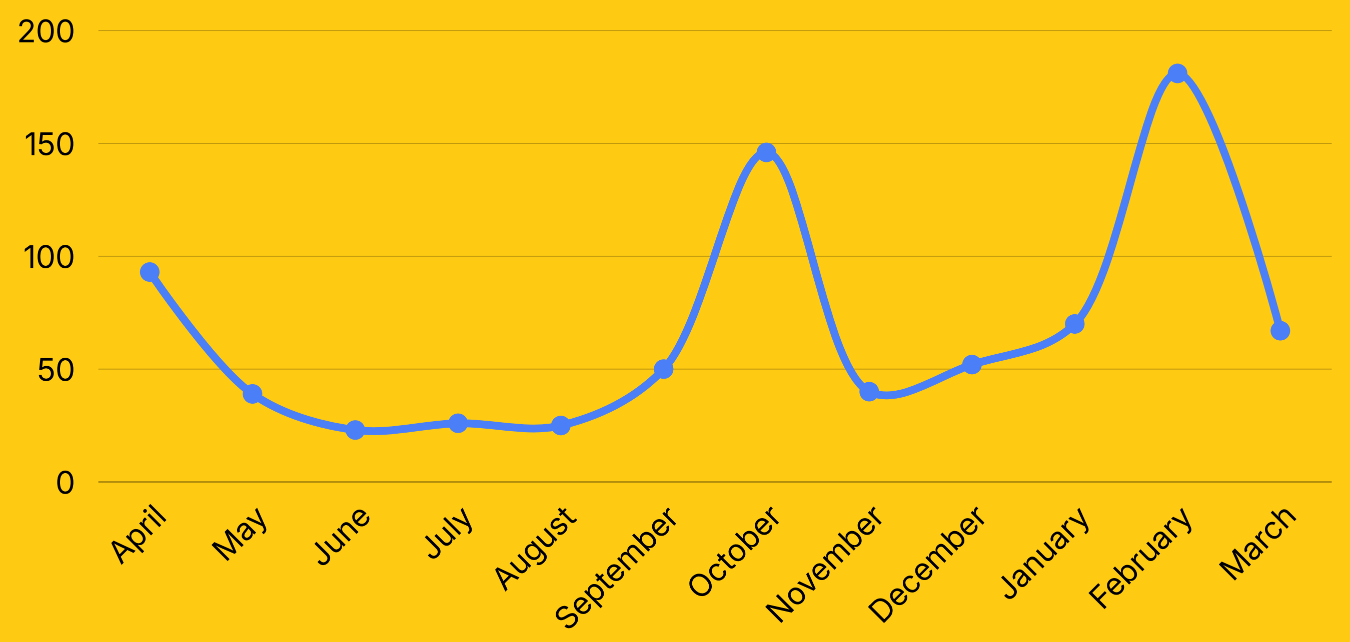
Temporal Profile

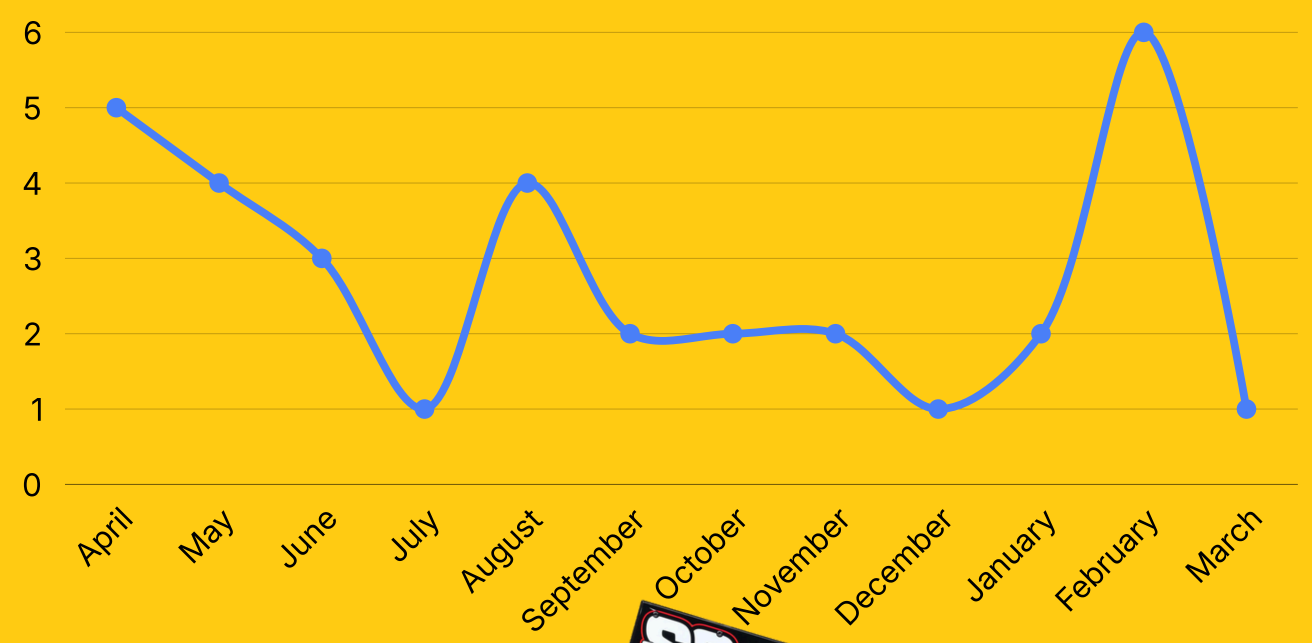
Real Data Training

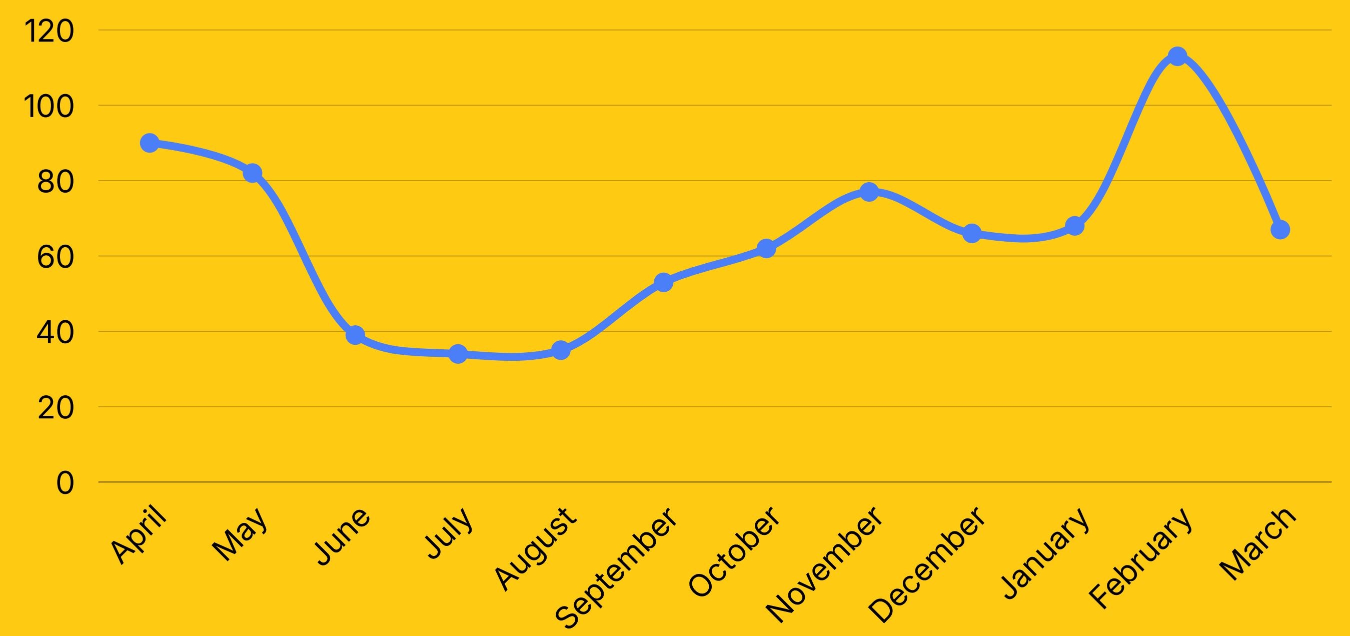
Grouping/Categorization

TEMPORAL PROFILE



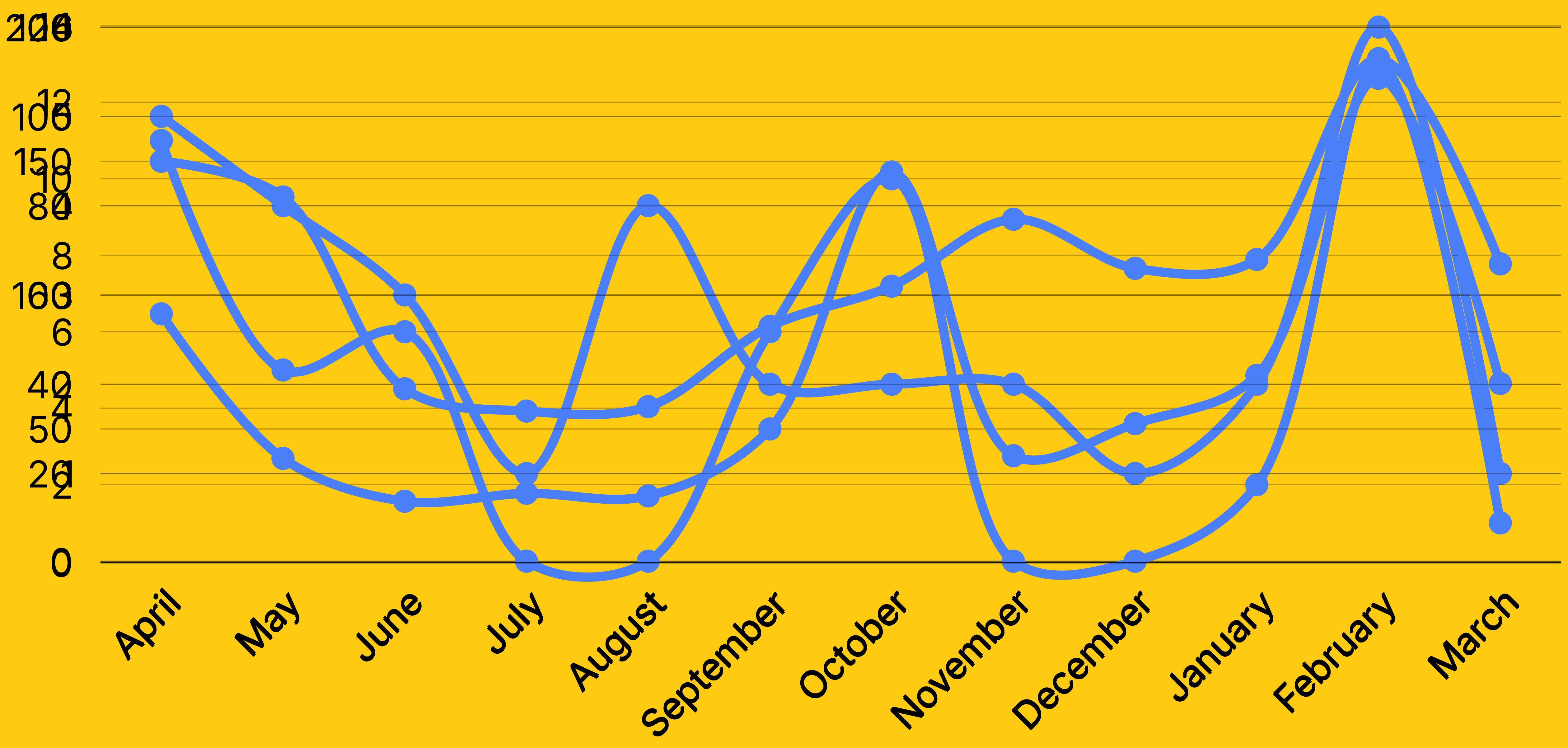






PARTS' ACTUAL DEMAND OVER THE YEAR

TEMPORAL PROFILE



REAL DATA FROM MULTIPLE SOURCES

Lack of actual data

Deep Research
- Reddit
- Online Reviews
- Youtube
- Public Posts on Social Media
- Blog posts, etc.

Compile into an Excel Sheet

Feed into the model

GROUPING/CATEGORIZATION

Multiple ways to Group them:

- Logical categorization - Tyres- Tyres, wheel hex, etc
- Correlational Grouping - Solely Based on Data
- Complimentary Grouping - Front bumper & suspension arm
- Use type - Electronics, Mechanical, Electrical

GROUPING/CATEGORIZATION

#	Sub-Type Name	Representative Parts
1	**Tires & Wheels**	Tires, foam inserts, wheel hex adapters
2	**Steering & Linkage**	Steering blocks, tie rods, bellcranks, servo savers
3	**Drivetrain & Gears**	Spur gears, pinion gears, differential assemblies
4	**Suspension**	Arms, hubs, shocks, springs, shock towers
5	**Chassis & Braces**	Main chassis, bulkheads, motor mounts, side guards
6	**Body & Cosmetics**	Body shells, wing mounts, decals, number panels
7	**Electronics**	Receivers, servos, motors, ESCs (Spektrum brand)
8	**Bearings**	Ball bearings (LOSA-prefix parts)
9	**Hardware**	Screws, clips, pins, fasteners (AXI/AXIC, ARA/ARAC)
10	**Shared Platform Parts**	TLR-prefix competition/racing crossover parts

RESULTS & IMPROVEMENTS

Metric	Legacy Forecast	Predictive Model
Total Part Demand — New SKUs	35,040 units	**42,043 units**
wMAPE (weighted forecast error)	**61.90%**	**24.90%**
Over-Forecast Bias Eliminated	98%+ systemic bias	Eliminated via category adjustment factors
Tire Under-Forecast Corrected	35% max planned rate	**56% corrected rate (+60%)**

RESULTS & IMPROVEMENTS

Fill Rate Metric	Value
Legacy Forecast Fill Rate	80.60%
Predictive Model Fill Rate	95.40%

RECOMMENDATIONS

- **Train with real data**
- **Fix Temporal Profile**
- **Iterate different grouping possibilities**
    - Create different Failure Cascades

FIX TEMPORAL PROFILE

- Train on as much training data as possible
- Start Testing on a single month - For eg. only for April
- If accuracy is achieved, test for April & May & so on
- If forecast is wrong — generate synthetic data
- Reiterate until required accuracy is achieved

FIX GROUPING METHOD

- Group/Categorize with different methods
- Start testing with the perfect Temporal Profile
- Record accuracy for all models
- Use the best one

FAILURE CASCADE ARCHITECTURE

Failure Type	Cascade Sequence	Demand Implication
Front-End Crash	(1) Front bumper flexes & absorbs → (2) Hinge pins & suspension arms snap → (3) Caster blocks and spindle hubs shear	Buying front bumper predicts arm demand within 1-2 weeks (trailing indicator)
Drivetrain Snap-Start	(1) Slipper clutch pads slip to protect gears → (2) Idler gear strips if slipper is locked → (3) Layshaft twists in extreme cases	Slipper pad demand is leading indicator of differential and layshaft demand
Differential Early Failure	Factory threadlocker insufficient → diff cover screws back out in first 2 battery packs → fluid loss → bearing contamination	25% launch-window demand spike for diff gear sets in M1; trailing bearing demand in M2-M3
Thermal Motor Failure	High ambient temps (July) + blue-groove clay tracks → motor exceeds 180°F → bearing failure accelerates	Electronics demand peaks July-August; flat rest of year. Requires seasonal index adjustment

USE THE BEST MODEL

Create a matrix of Temporal Profiles & Groupings

Test all the possibilities

Record accuracy for all models

Use the best one

RECOMMENDATIONS

Pull Strategy for months with very low demand

- Hold (very small number) orders
- Saving investment for that period of time
- Increased value for Horizon (assumption)
- Trade-off being reduced customer service level standard

Increase Prices of parts just before Launch

- Sometimes forces consumer to make a choice
- Buy a new car or invest in the old one
- Horizon earns extra in both the cases

THANK YOU!

We are happy to take questions now

OUR MODEL

Random Forest Regressor

Gradient Regressor

K-Means